

NxM Templates using Run 4 Ge Activation Data

K-line event selection · two-exponential fits · 1x1 and NxM (PCA) templates

Future step: three, four exponential fits

Zhiheng Li — July 2026

Table of Content Dataset Data Cleaning Pulse fit Template generation
Future steps

Ge activation dataset - R4

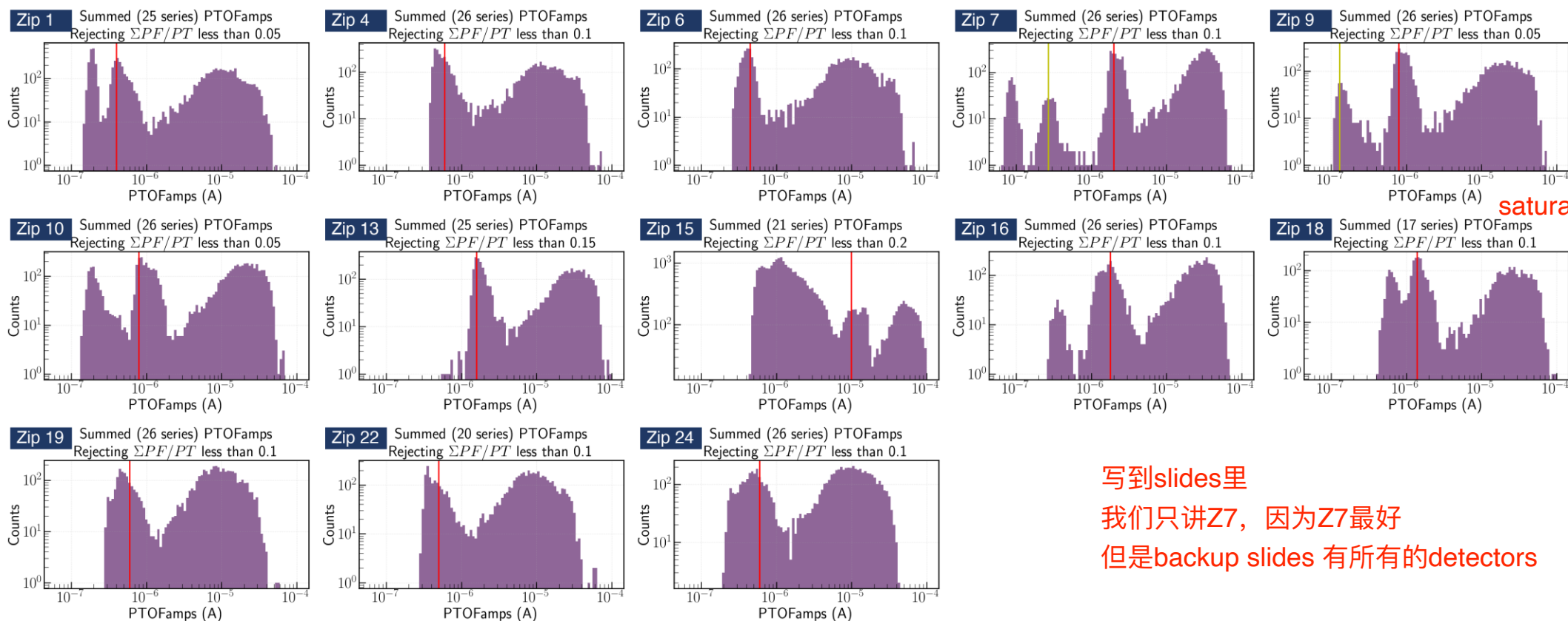
From the Ge activation K line to an event sample

Ge Activation Data — Ops Shift 260612

Credit: Prof. Tarek Saab — SuperCDMS Confidential

~~Goal: one clean, minimally biased pulse population per detector × channel~~

- Per detector, Prof. Saab's study marked the 10.37 keV K-line position in PTOFamps — the red line in each panel below
- Our event selection: a PTOFamps window bracketing the red line position $\times/\div 1.35$ (a rough, somewhat arbitrary eyeballed choice); deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata “Pickle file”; 好处就是重复使用
- 13 detectors (zips) × 27–30 series ~~full raw traces cached for easy access~~ 大概有多少个event



saturated events

写到slides里

我们只讲Z7，因为Z7最好

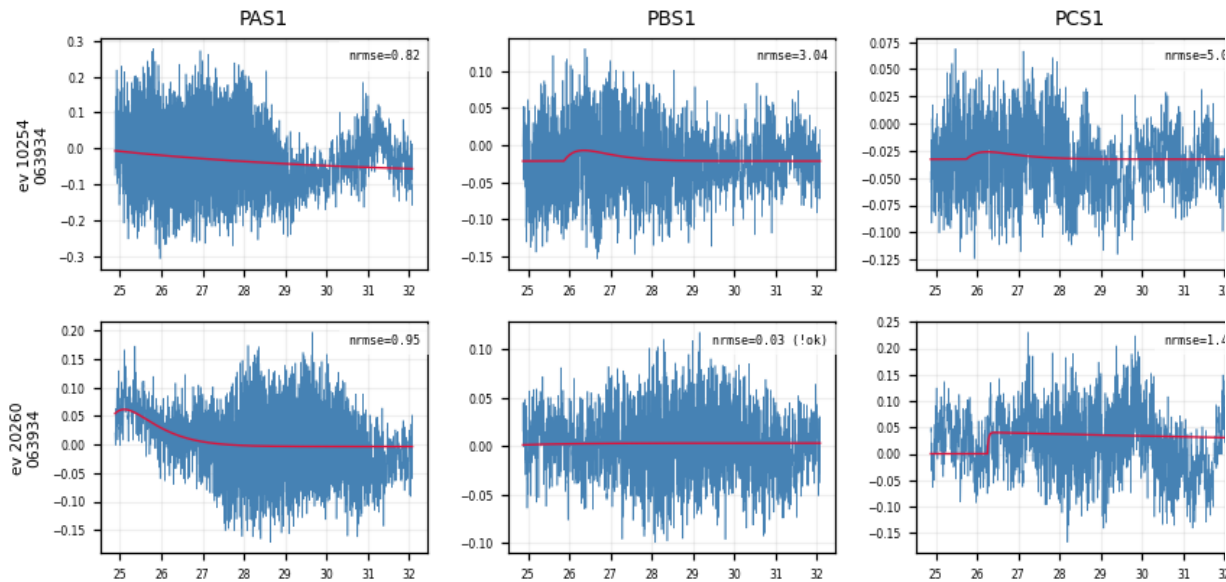
但是backup slides 有所有的detectors

Fit quality at a glance — event × channel grids

- ~~One row per event, one column per channel: the low-passed trace (blue) with the fitted pulse model (red) overlaid~~
- ~~A real event fits well in every channel at once; a noise trigger fails in every channel~~

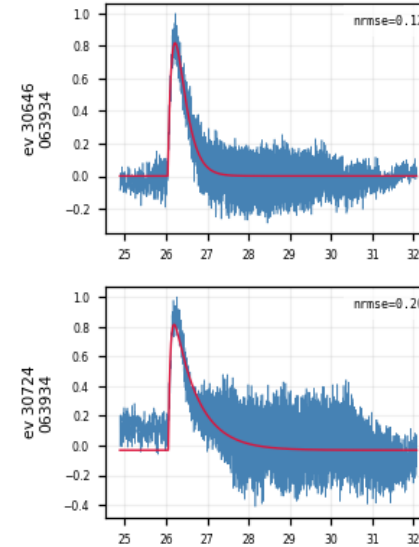
Visual comparison

Noise

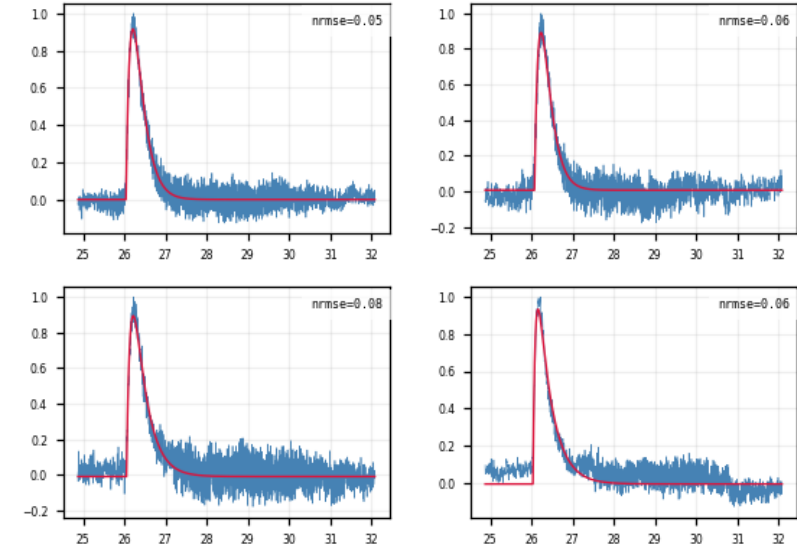


Noise triggers (Z7): ~~the fit fails in every channel at once~~

PAS1



Real pulse



K-line events (Z7): ~~consistent good fits across channels~~

Per-trace algorithm: low-pass → normalize → fit → align

1. **Low-pass** — 100 kHz Butterworth how baseline is calculated?
2. **Normalize** — subtract the baseline, scale the trace to unit peak
3. **Fit** — two-exponential model, ~~all 5 parameters free (including the pretrigger t_0):~~

$$y(t) = A[e^{-(t - t_0)/\tau_{\text{fall}}} - e^{-(t - t_0)/\tau_{\text{rise}}}] + b$$

t_0 = pretrigger, free within 16050 ± 3000 samples (τ_{rise} , τ_{fall} : rise / fall time)

Fit parameter definitions

4. **Two quality checks per trace** (only recorded, not cut yet):

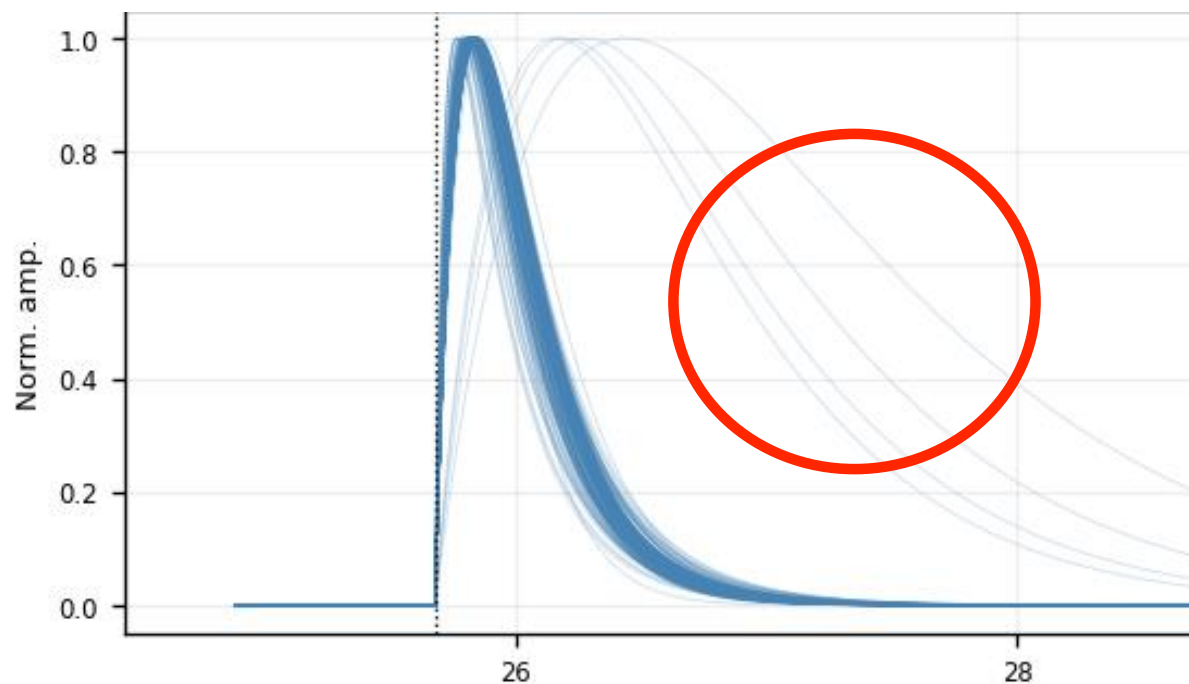
- **fit_ok** — amplitude is positive, and the pulse rises faster than it falls calculated and recorded
- **NRMSE** — RMS of the fit residual ÷ pulse peak (smaller = better fit)

risetime -> L/R, electrical property of the circuit
falltime -> C/G, thermal property of the detector

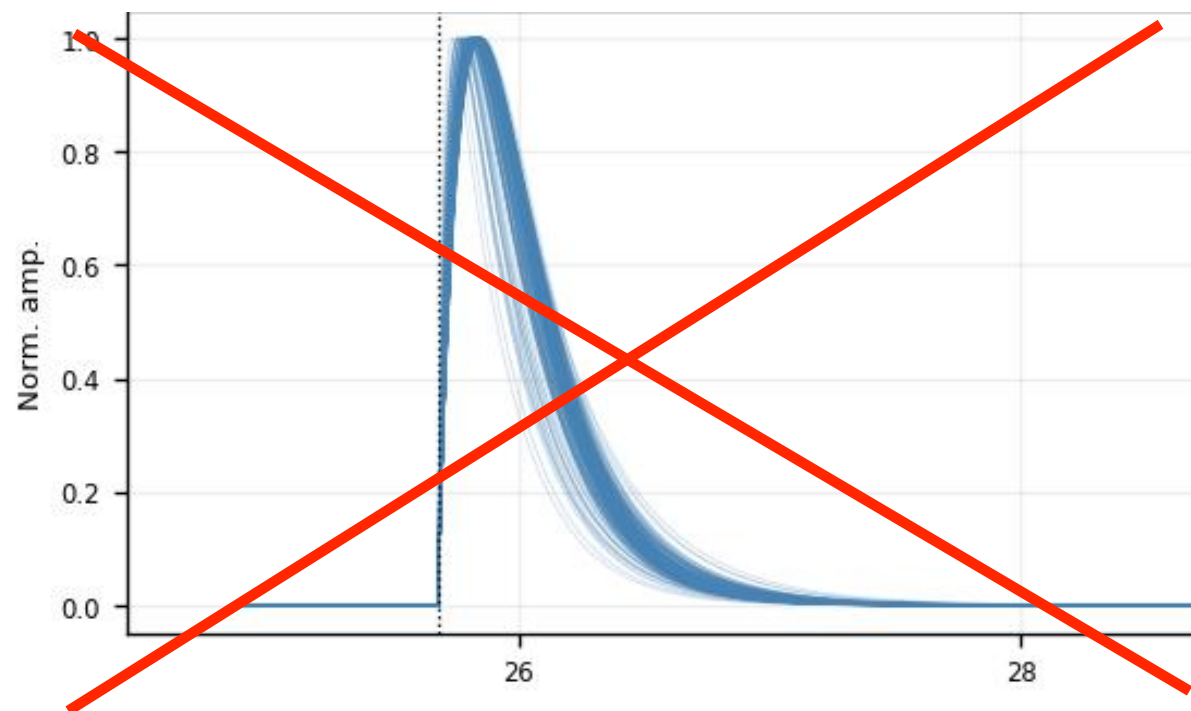
5. Align ~~— shift each trace to a common pretrigger~~

- Shift the measured trace by (fitted pretrigger - 16050) ~~— a pure translation. Drawn at the common pretrigger, the fitted curves stack into one shape family: — —~~

只放一个no cut 的图,
include number of events



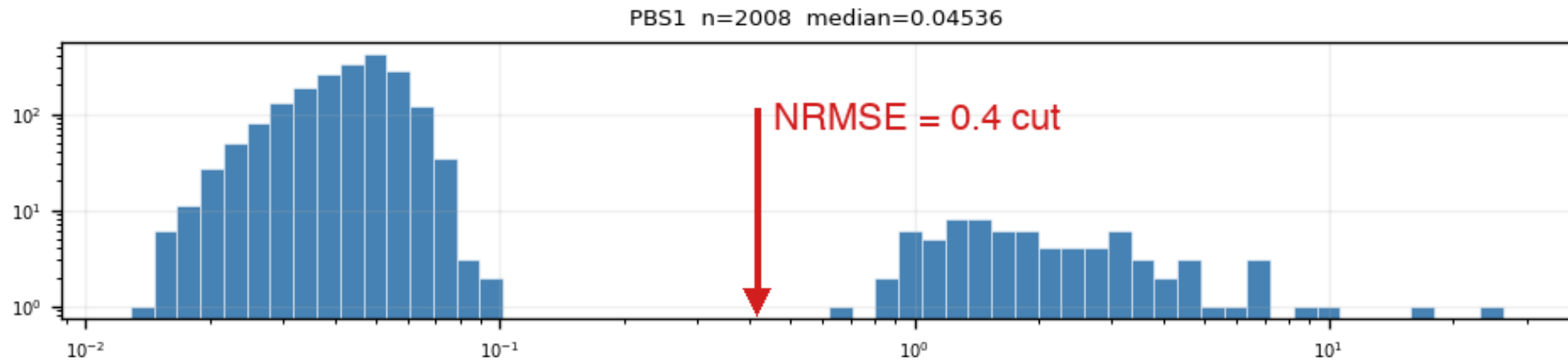
All fit_ok fitted curves (Z7 PBS1) — no cut



After $NRMSE \leq 0.4$ (Z7 PBS1) — one tight shape family

Quality cut 1: $\text{NRMSE} \leq 0.4$ — where the number comes from

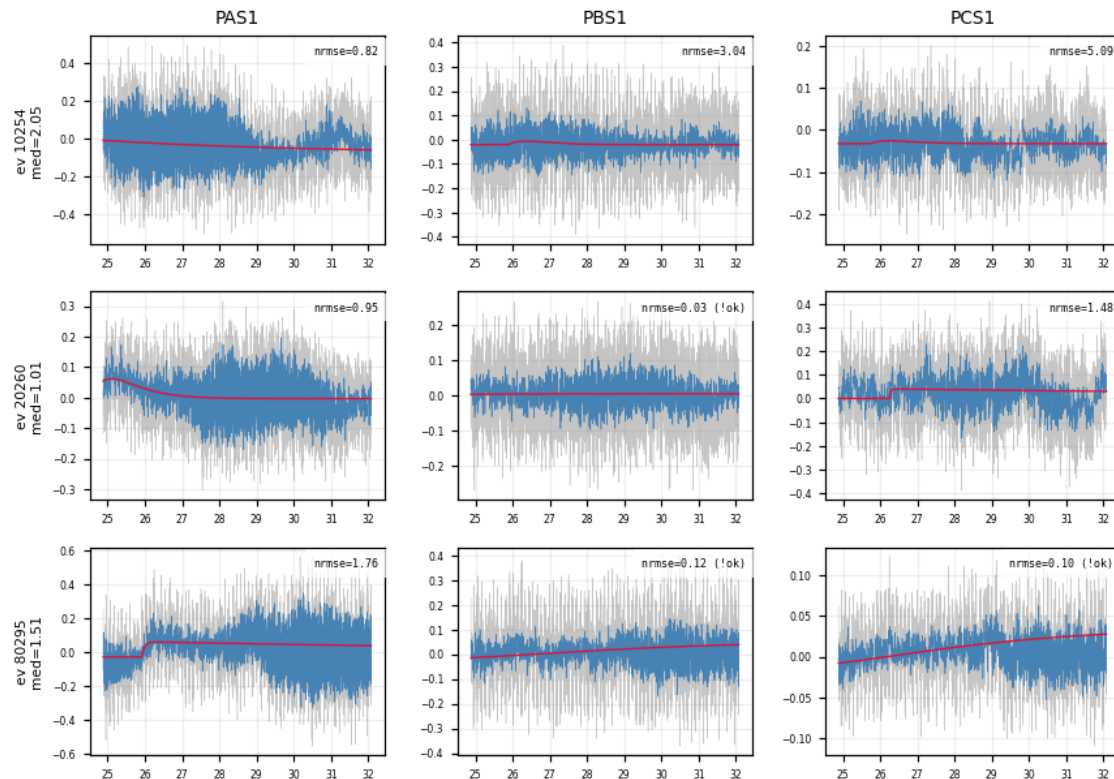
- NRMSE of fit_ok events is bimodal: good fits (median ≈ 0.05 – 0.1) vs noise triggers (≈ 1 – 2), valley at ≈ 0.4 – 0.5
- The cut sits in the valley: eyeball read off the distribution
- Worse detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): same bimodal picture, noise peak dominates



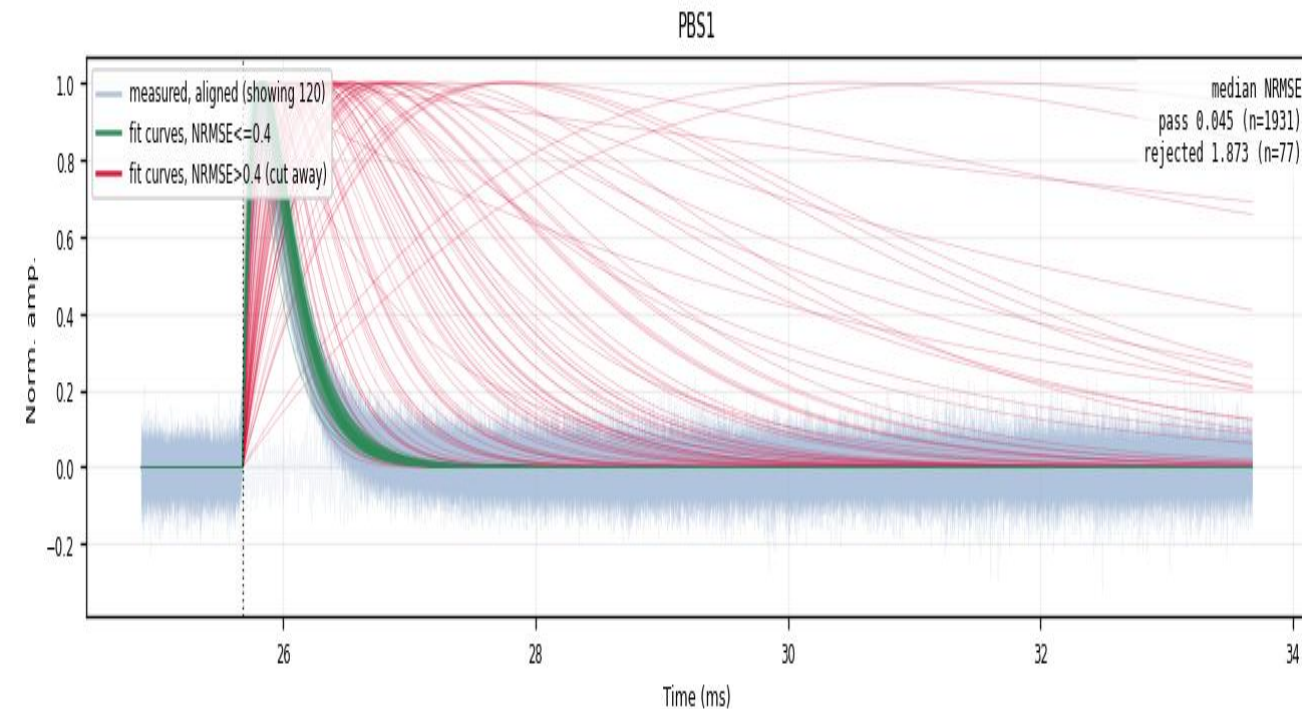
Z7 PBS1: two clean populations, log-log axes

The rejected population is noise — verified in the raw traces

- Example of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse
 - the cut removes noise triggers, not physics



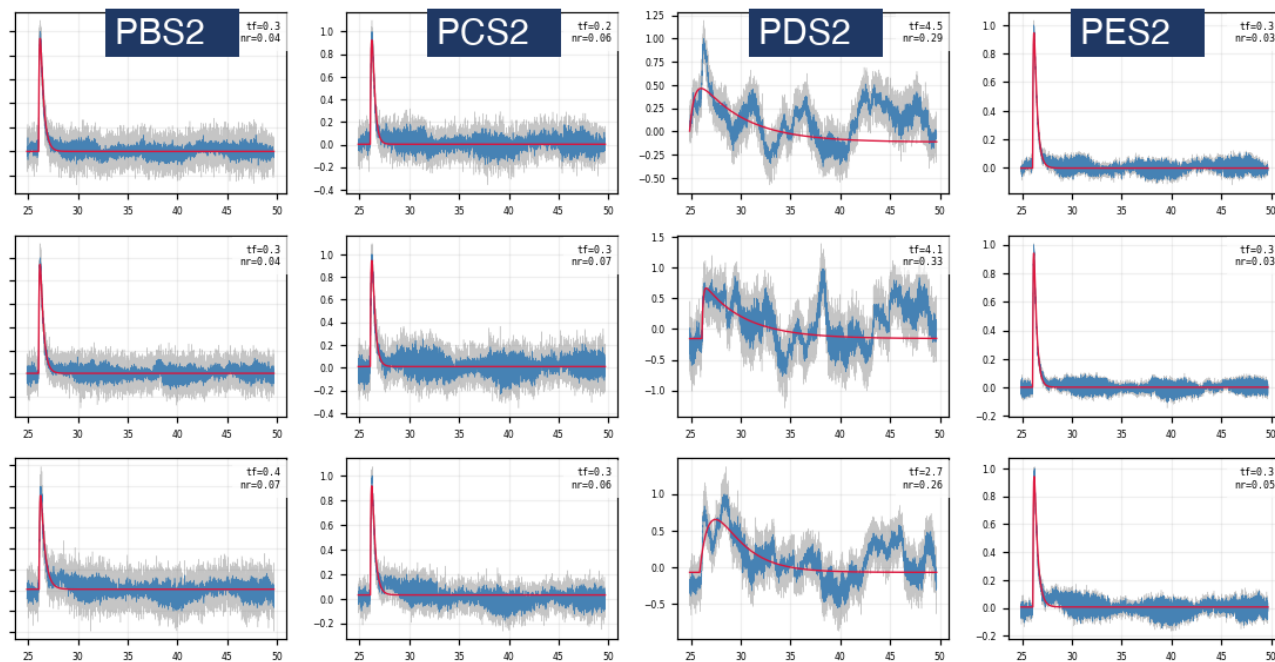
Z7: NRMSE-rejected events — raw traces are noise



Z7 PBS1 — passes the cut (kept) — cut away = rejected
faint blue = measured traces 1931 kept / 77 cut (~3.8%)

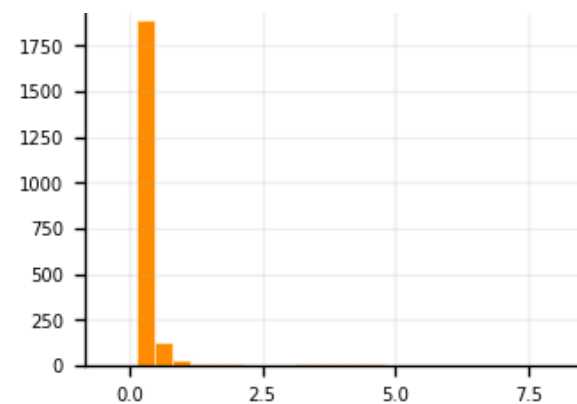
Slow-fall tail in one particular channel

- The post-cut plot still shows slow-fall tails → sample them: $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{fall}} > 1.5$ ms, 10 random events, raw vs fit drawn in all 12 channels
- The sampled events are real pulses. Their extreme fall times trace to one channel: only PDS2 swings (τ_{fall} median 0.51 ms vs ≈ 0.25 ms elsewhere)
- Conclusion: no τ_{fall} cut — slow-fall events passing NRMSE stay in

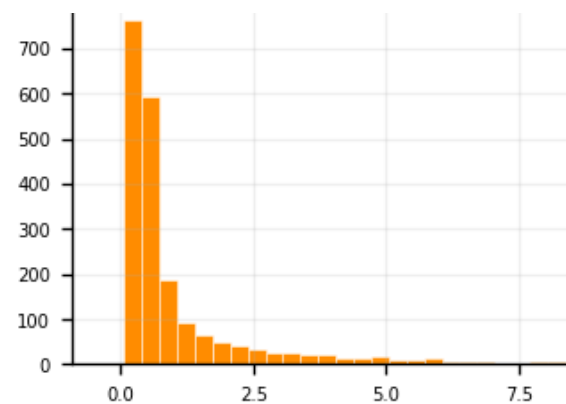


Z7, 3 sampled slow-fall events: normal in PBS2/PCS2/PES2, swings only in PDS2

Fitted τ_{fall} distribution, same 0–7 ms axis:



Z7 PAS1 — normal channel: narrow
(median 0.25 ms)



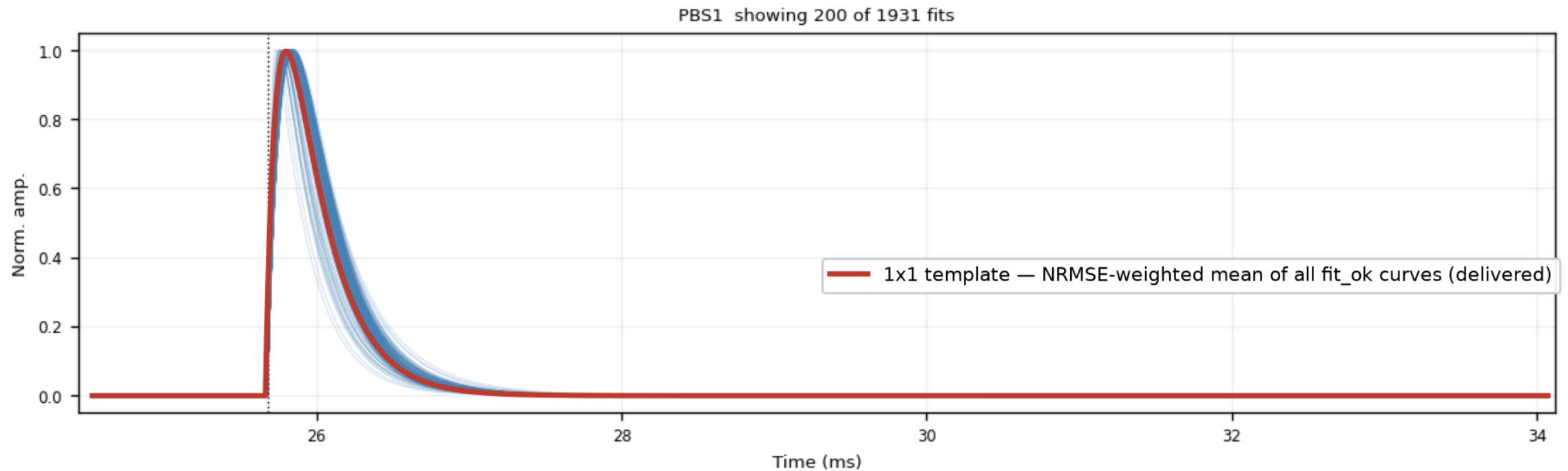
Z7 PDS2 — bad channel: broad tail
(median 0.51 ms)

can not make useful templates
analysis/software hard to fix
need to fix noise
temporary solution: 用PDS1 templates代替 PDS2

Results: 1x1 Templates

- NRMSE-weighted mean of the fit_ok 2-exp curves
- Smooth & noise-free by construction

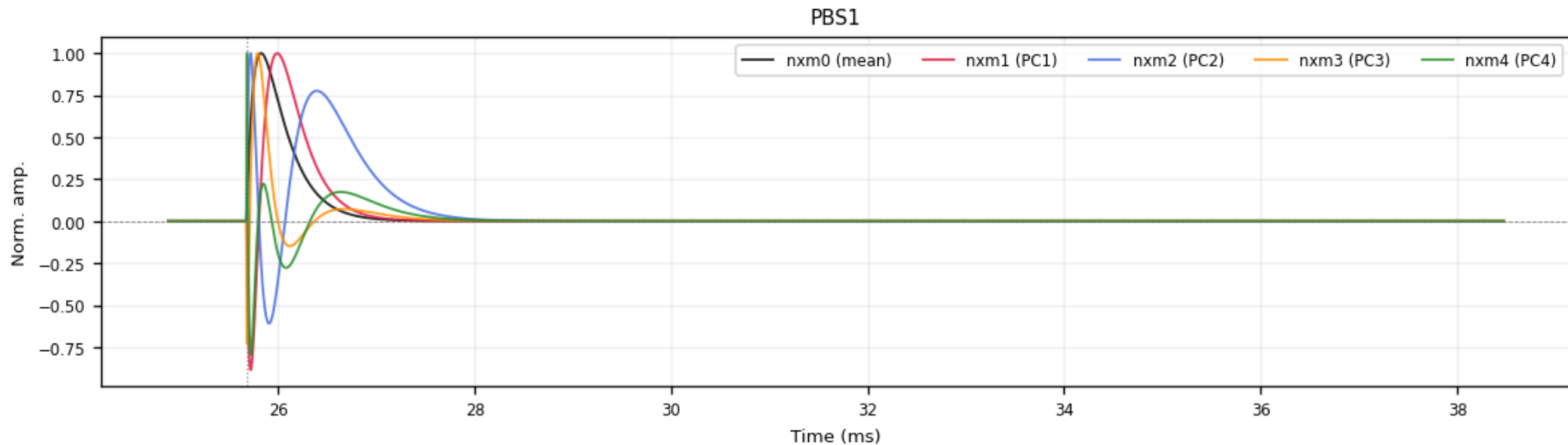
再加一个没有weighted的简单平均
再加一个NxM0



Z7 PBS1 — blue: the fitted curves (200 of 1931 drawn); red: the ~~delivered~~ 1x1 template, ~~the NRMSE-weighted mean of all fit_ok curves, read back from the ROOT file~~

Results: NxM Templates

- channel-specific PCA over ~~the clean~~ fitted curves ($\text{fit_ok} + \text{NRMSE} \leq 0.4$ ~~+ $\tau_{\text{rise}} \leq 0.3$ ms~~)
- nxm0 = mean shape; nxm1–4 = principal components; real pulse = $\sum_i \text{amp}_i \cdot \text{nxm}_i$
- PC1 + PC2 $\approx$ 96–98% of the shape variance



Z7 PBS1: nxm0 (mean) + nxm1–4 (PCs) — the oscillating components encode rise/fall-time variation

Template file

- Two template sets for all 13 detectors: 1x1 (2-exp weighted) + NxM PCA
- In cdmsbats_config feature branch, ready for merge

Future Steps

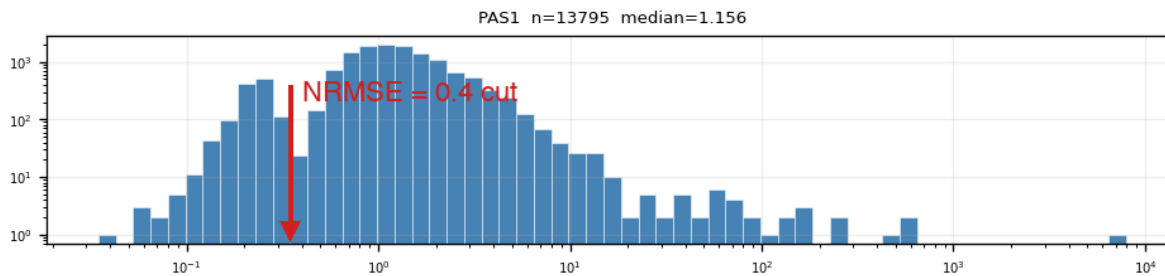
3, 4, expo fit

exercise NxM processing

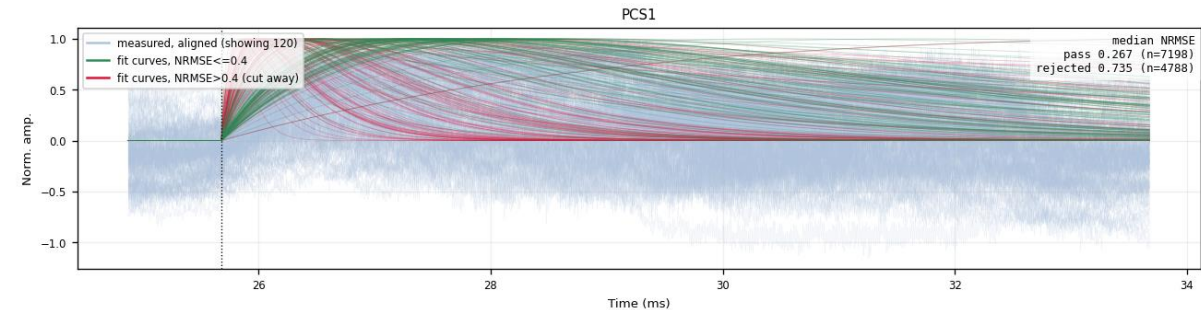
other detectors - worse noise, added a few more cuts, in the backup slides,如果大家有兴趣, 我可以讲一讲backup slides

Backup — worse detectors (example: Z22)

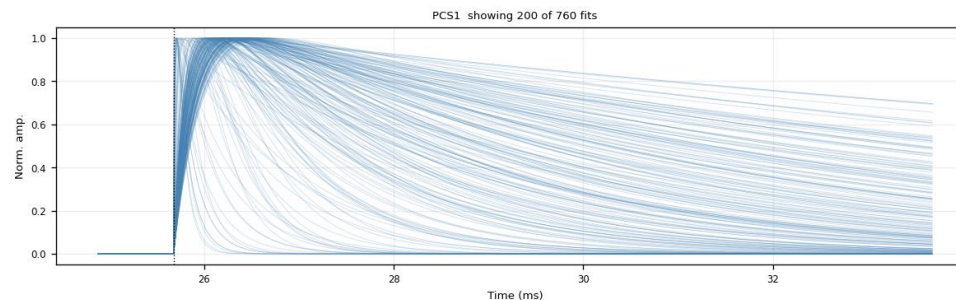
- Z1 / Z4 / Z6 / Z18 / Z19 / Z22 / Z24: K-line sits inside the noise population — the PTOF window admits a noise-dominated mixture
- Same bimodal NRMSE picture, noise peak dominates; however, the same 0.4 cut digs the real pulses out — Z22 PCS1: 7198 kept / 4788 cut (~40%)
- $\tau_{\text{rise}} \leq 0.3$ ms (after NRMSE): removes 55–74% on worse zips (Z22: 71%) vs 1.6% on Z7



Z22 PAS1: NRMSE histogram — noise dominates



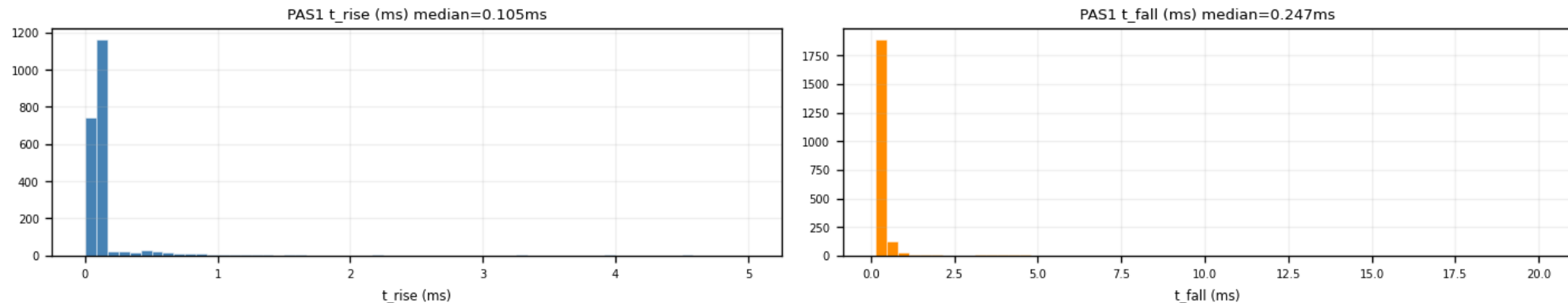
Z22 PCS1: kept (green) vs cut (red)



Z22 PCS1 after $\text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3$ ms — final shape family (760 fits)

Backup — the $\tau_{\text{rise}} \leq 0.3$ ms ceiling (PCA input)

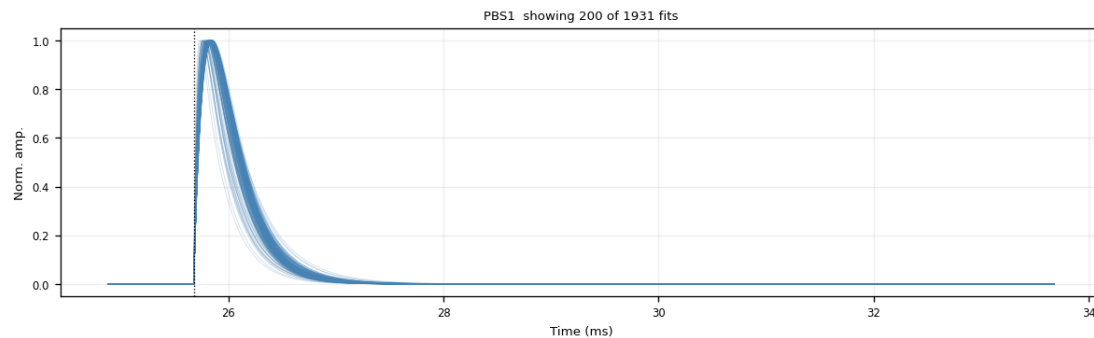
- Slow baseline drift survives NRMSE (a slow 2-exp hugs it) → mimics a slow rise
- Fast pulses at $\tau_{\text{rise}} \approx 0.1$ ms (p90 ≈ 0.15 ms) → ceiling at 0.3 ms blocks the drift tail
- Removed after the NRMSE cut: Z7 1.6% (quiet 2–6%) — weak 55–74% (Z22 71%) — all zips 54%
- Trade-off: trims the very slowest genuine pulses → decision pending



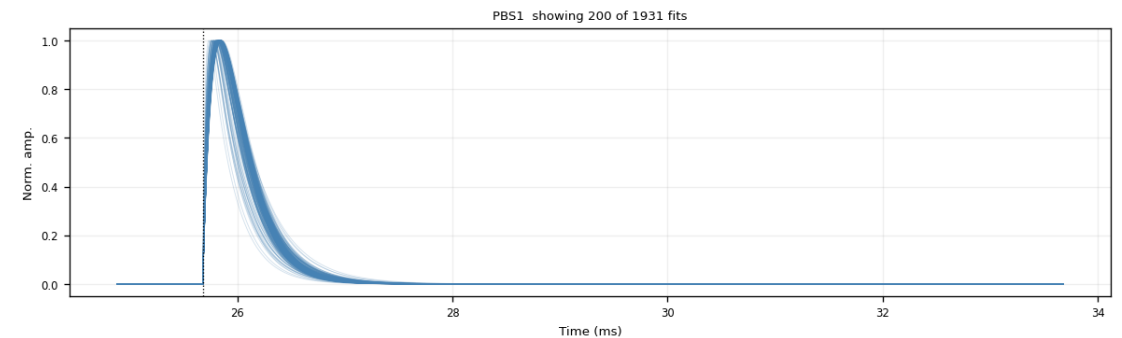
Z7 PAS1: fitted τ_{rise} (median 0.105 ms) and τ_{fall} distributions

Backup — τ_{rise} cut: a good vs a bad channel (Z7)

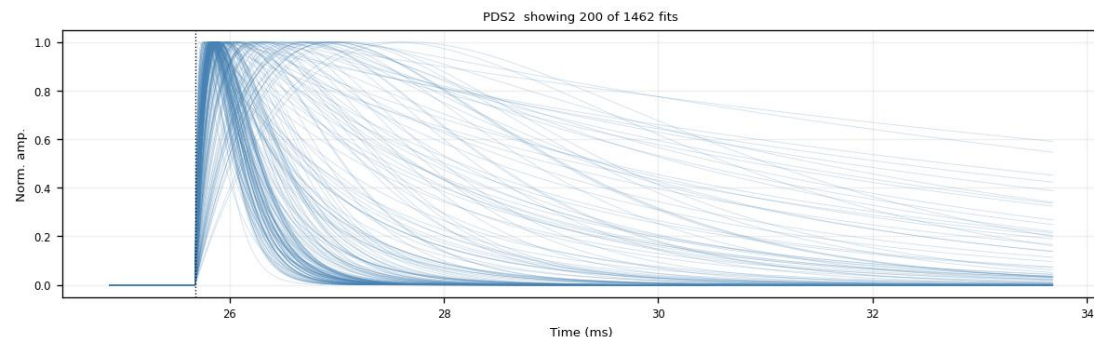
- Good channel PBS1: tight clean bundle — the $\tau_{\text{rise}} \leq 0.3$ ms cut removes $\sim 0\%$ (nearly free on quiet channels)
- Bad channel PDS2: broad, messy — the channel itself is bad; the cut trims the slow-rise curves, 1462 $\rightarrow$ 1154 (21%)



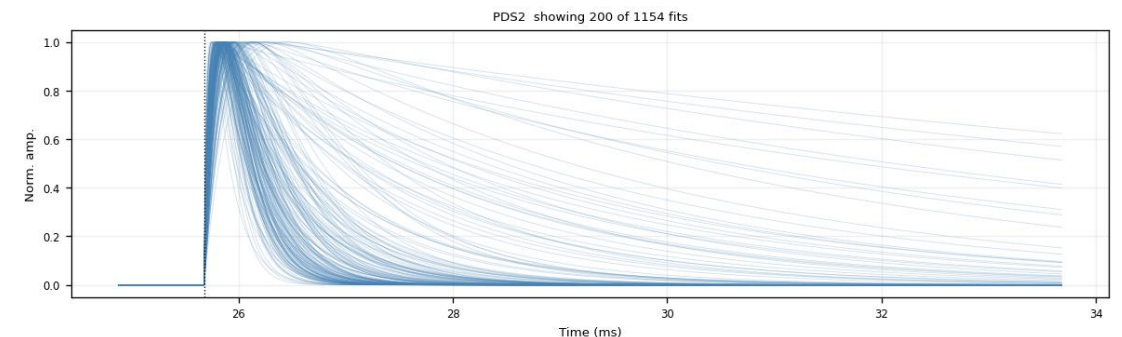
PBS1 (good) — $\text{NRMSE} \leq 0.4$



PBS1 (good) — $+\tau_{\text{rise}} \leq 0.3$ ms (unchanged)



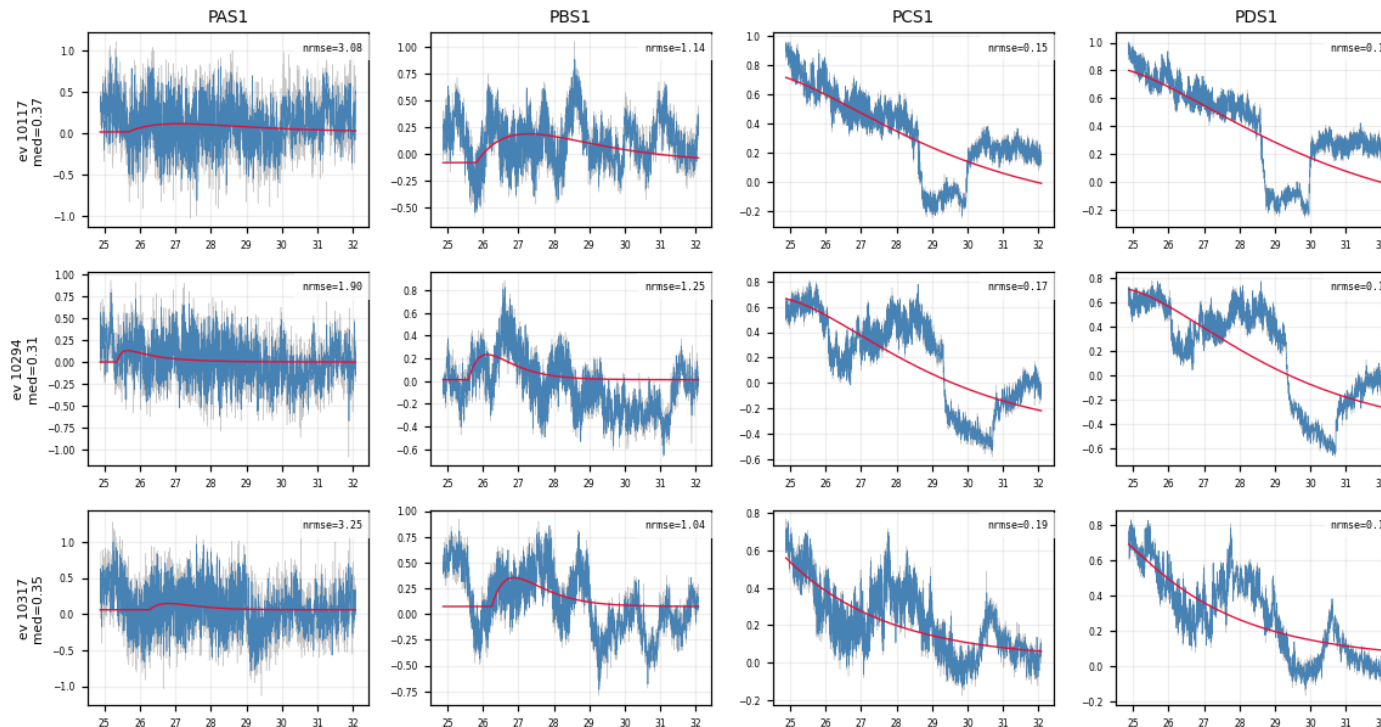
PDS2 (bad) — $\text{NRMSE} \leq 0.4$



PDS2 (bad) — $+\tau_{\text{rise}} \leq 0.3$ ms (slow risers trimmed)

Backup — what the τ_{rise} cut removes: noise NRMSE missed (Z22)

- Exactly the events that PASS $\text{NRMSE} \leq 0.4$ but are REMOVED by $\tau_{\text{rise}} \leq 0.3$ ms — the noise 0.4 let through, 0.3 catches
- PCS1 / PDS1 pass at $\text{NRMSE} \approx 0.15$ (a slow 2-exp hugs a slow baseline drift) — but the raw trace has no clean fast pulse
- Trade-off: the same cut also trims genuine slow-rise pulses — decision pending



Z22: 3 events $\times$ PAS1/PBS1/PCS1/PDS1 — pass $\text{NRMSE} \leq 0.4$, removed by $\tau_{\text{rise}} \leq 0.3$ ms; PCS1/PDS1 are pure slow drift